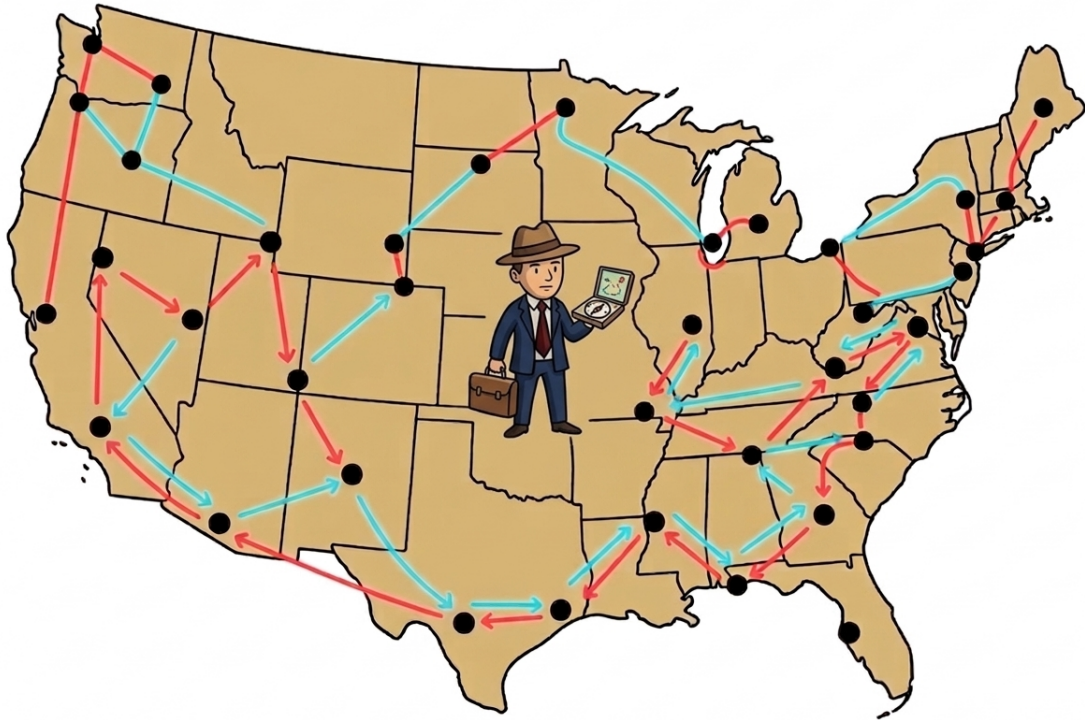


Solving the Traveling Salesperson Problem (TSP) using Evolutionary Computation



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Abstract

This study addresses the Traveling Salesperson Problem (TSP) using Genetic Algorithm (GA) techniques. A comparative analysis is conducted between a baseline approach (OX1 Crossover and Swap Mutation) and an improvement proposal oriented toward adjacency preservation (Edge Recombination Crossover and Inversion). The results demonstrate that transitioning from an order-based logic to an edge-based logic improves convergence efficiency by up to 1000% on the 48-city dataset.

1 Introduction

The Traveling Salesperson Problem (TSP) is a classic combinatorial optimization challenge classified as NP-hard. This project implements an Evolutionary Computation environment in Python to minimize the traveled distance in a 48-city route, simulating natural selection processes: crossover, mutation, and survival of the fittest.

1.1 Study Objective

The main objective is to reach an optimal solution defined by a total distance traveled $D \leq 35,000.00$ units, analyzing the algorithm's sensitivity to variations in population critical mass and mutation rate.

2 Experimental Setup

The experiment was developed in a *Jupyter Notebook* environment using standard Python libraries (NumPy, Random, Matplotlib). The fundamental hyperparameters are detailed in Table 1.

Table 1: Experimental Setup Parameters

Parameter	Description
population_size	Number of individuals (routes) in each generation.
num_generations	Maximum iteration limit (set to 2000).
mutation_rate	Probability of genetic alteration for diversity.
Stopping Criterion	Distance $\leq 35,000.00$ or iteration exhaustion.

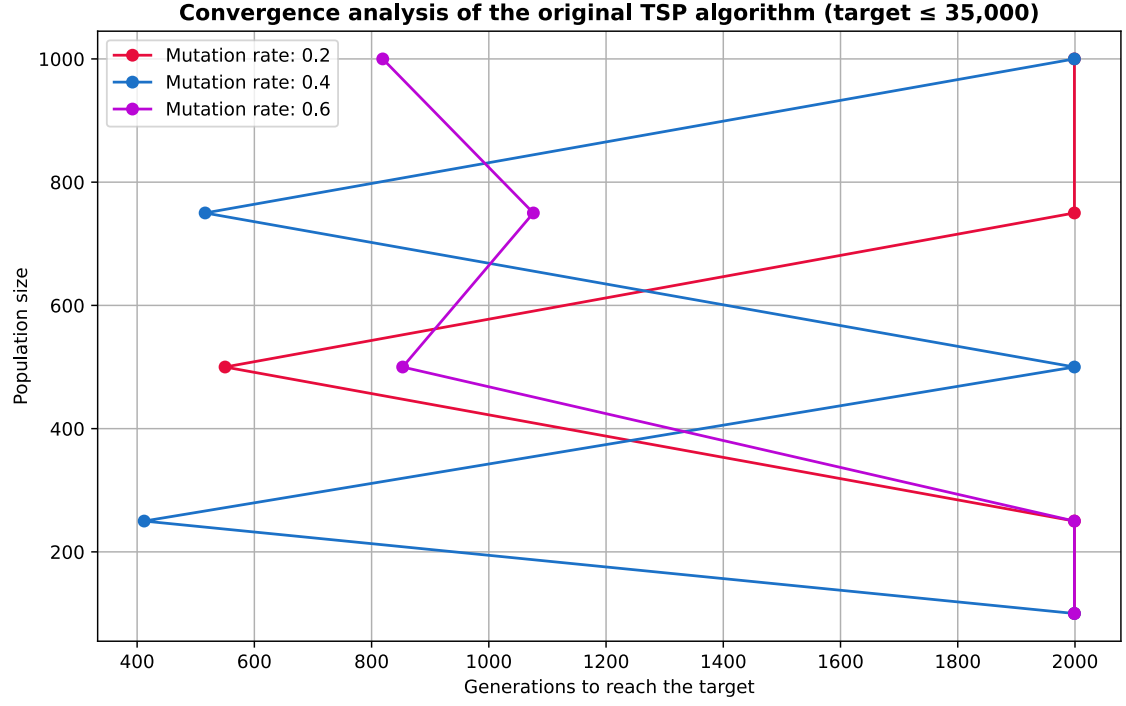
3 Parametric Study: Original Algorithm

Convergence was analyzed by varying the population in the $[100, 1000]$ range and mutation in $[0.2, 0.4, 0.6]$. The baseline model uses:

- **Order 1 Crossover (OX1):** Preservation of relative order via toroidal logic.
- **Swap Mutation:** Random swapping of two alleles (cities).

3.1 Convergence Observations

Statistical analysis reveals a highly stochastic nature. Low mutation rates (0.2) were identified as prone to getting trapped in local optima due to an excessively conservative character. Curiously, the 0.6 rate showed greater success by introducing the necessary chaos to explore new regions of the search space, compensating for the inefficiency of the *Swap* operator in long routes.



4 Improvement Proposal: Adjacency Approach

Upon detecting the instability of the OX1 model, a paradigm shift toward edge preservation is proposed.

4.1 Edge Recombination Crossover (ERX)

The ERX operator constructs an *Edge Table* to prioritize inheriting direct connections. The smart selection of the next node is based on the principle of the fewest remaining edges, ensuring that the offspring inherits between 95% and 99% of the parental edges.

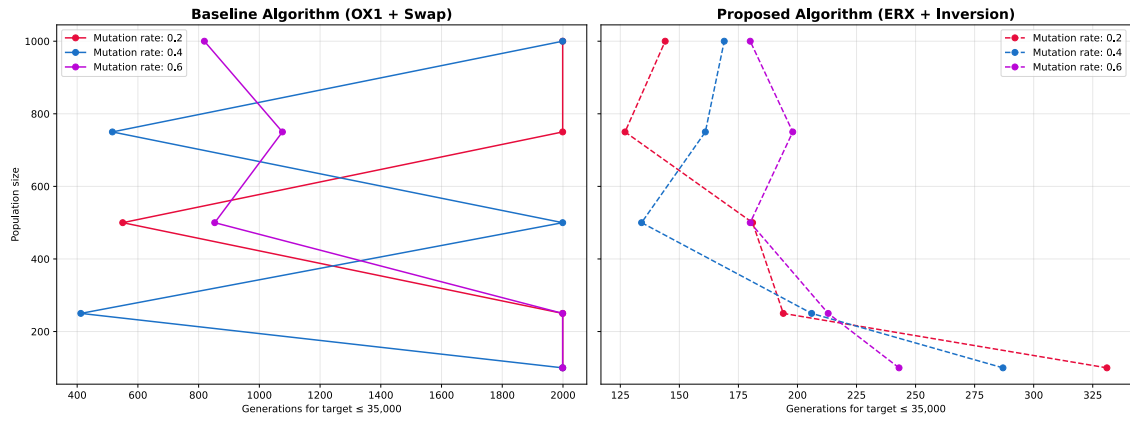
4.2 Inversion Mutation

Unlike *Swap*, which breaks up to 4 connections, inversion only alters 2, acting as a local search optimizer that untangles route segments without destroying the global structure.

5 Results and Conclusions

The comparison between the baseline model and the proposed one (*ERX + Inversion*) yields the following technical conclusions:

1. **Convergence Efficiency:** The proposed model reaches the target fitness between generations 125 and 330, compared to the 2000 required by the baseline model.
2. **Robustness:** The proposal demonstrates independence from critical mass; even with a population of 100 individuals, the algorithm is capable of converging, representing a 90% computational saving.
3. **Stability:** The convergence curves of the proposed model are predictable, indicating minimal reliance on chance once a high-quality adjacency inheritance is established.



References

- [1] *Edge Recombination Crossover (ERX)*. Resource used for the pseudocode approach and adjacency logic of the proposed improvement algorithm. Available at:
<https://k78ma.github.io/quartz/4B/ECE-457A/Edge-Recombination-Crossover>